

TeleCon4ESP32 User Guide

Phone app for Control Panel and RC Vehicle

For people who use the Android application

Firmware authors: see the Control Protocol PDF

Version 1.0 2026

Abstract

This guide is for the person holding the phone. It shows how to install TeleCon4ESP32, flash a starter ESP32 sketch, connect, and drive **Control panel** or **RC Vehicle**. Byte formats, UUIDs, and checksums live in the Control Protocol document. Per-sketch pinouts and Serial banners live in that sketch's user guide.

1 What TeleCon is

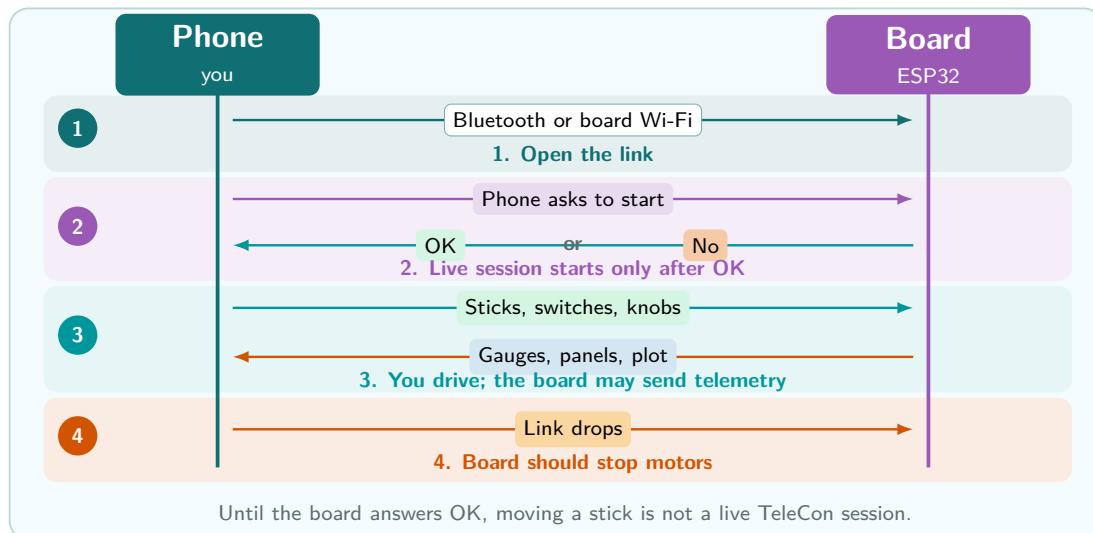
The phone is always the remote. The board (ESP32 DevKit, ESP32-CAM, or any MCU that speaks the contract) is the machine.

Two modules ship today:

Module	Access	What you do with it
Control panel	FREE	Dual sticks, switches, knobs, gauges, and plots for any ESP32 project.
RC Vehicle	PRO / coins	Driving cockpit plus optional ESP32-CAM video.

Control panel — free RC Vehicle — Pro Video is a separate Wi-Fi link

Both modules share the same control session. Camera video (HTTP `/stream`) is a second link: join the board's Wi-Fi for pictures; Bluetooth or TCP carries sticks.



2 What you need

- An Android phone (Android 12 or newer is simplest for Bluetooth).
- USB cable and a computer with Arduino IDE (to load the starter sketch).
- An ESP32 DevKit for the first session. An ESP32-CAM is optional and only needed for video.

Stay on **Default** and **Classic Simple** until that path works. **Advanced** (Binary, BLE, DevKit Wi-Fi, CAM Binary) is for later.

3 Safety

- Treat sticks as live once the app shows **Connected**. Wheels, servos, and ESCs can move.
- Until the board answers OK, on-screen motion is not a live session even if Android says Bluetooth is connected.
- If the phone drops the link, official sketches fail-safe (motors stop). Confirm that on your own firmware before a vehicle leaves the bench.
- One phone at a time. Close other apps that hold the same Bluetooth or Wi-Fi link.

4 First session

Goal: in about ten minutes, move a stick on **Control panel** and see the board react (Serial Monitor echo, or a motor if you already wired one).

Before you start. Phone charged. ESP32 powered from USB. Arduino IDE installed. You do not need Pro or coins for this path.

1 — Install the app

Install TeleCon4ESP32 from Google Play (or your debug build). Open it. Grant Nearby devices / Bluetooth when Android asks. On Android 11 and below, Location permission is still required for scanning; the app does not need Location to stay on.

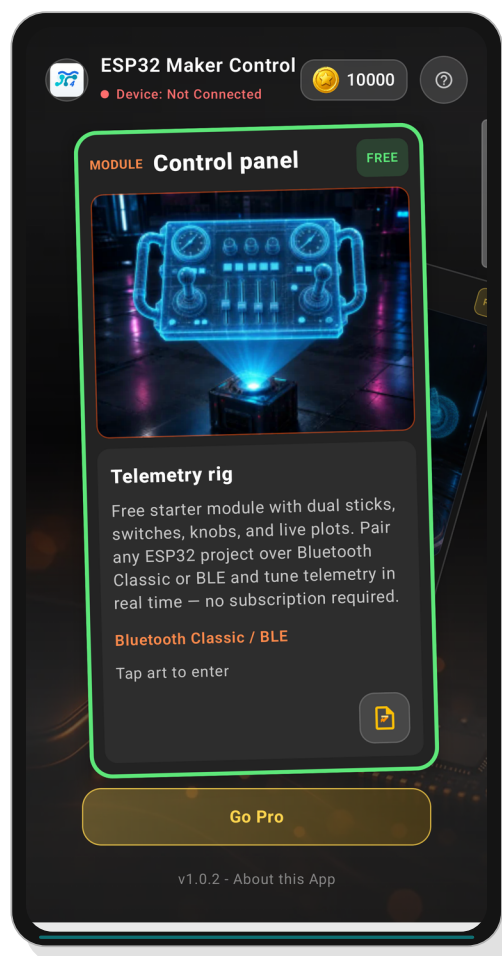


Figure 1: **S01.** Home screen after first launch. Show the top bar (connection status Not connected, optional gold PRO, coin chip, Help), the two module cards Control panel (FREE) and RC Vehicle (PRO), and the Codes icon on a card. Do not cover labels. English UI.

Portrait, light theme, not connected ■ s01_home.png

2 — Get the starter sketch

On Home, tap the docs icon on **Control panel** (or open **Codes and Documents** from that module). Share or save the DevKit Classic ZIP, copy it to the computer, and open the `.ino` folder in Arduino IDE.

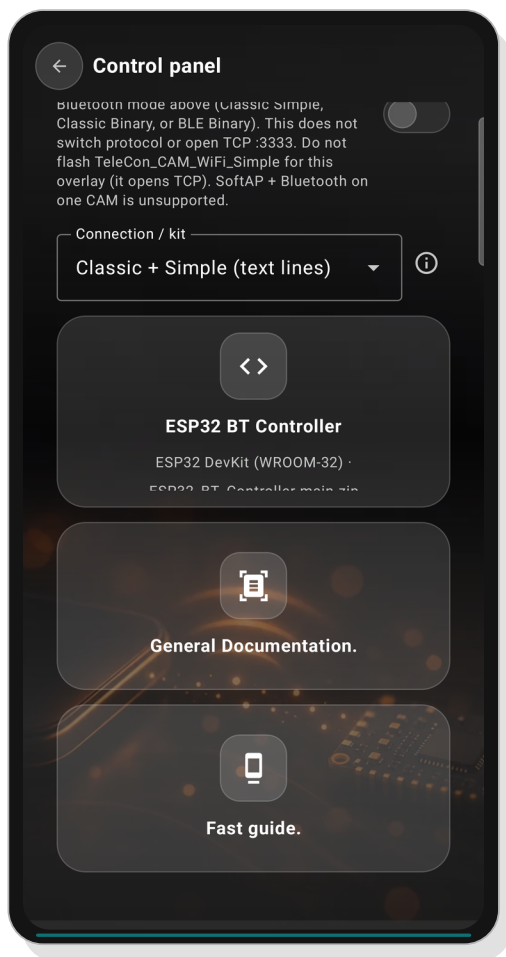


Figure 2: **S02**. Codes and Documents for Control panel. The general user-guide PDF, the fast guide, and the ESP32 Classic ZIP row must all be visible. English UI.

Portrait, light theme ■ s02_codes.png

Flash **TeleCon Classic Simple**. Board and Port under Tools must match your USB device. Serial Monitor at 115200. Press RESET. Wait for the ready banner, then the board is advertising ESP32-TC-RC-BT-Simple.

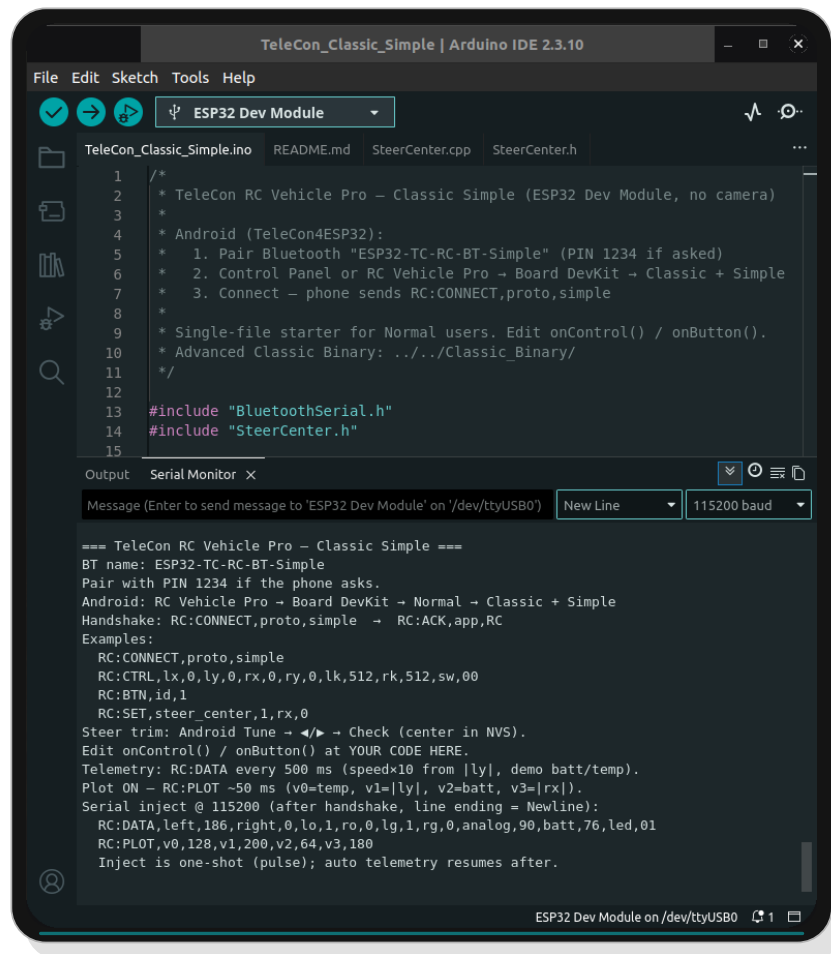


Figure 3: **S02b**. Serial Monitor at 115200 after RESET. Classic Simple ready banner and BT name ESP32-TC-RC-BT-Simple visible. Crop to the Serial window.

Computer, Arduino IDE Serial Monitor ■ s01_serial.png

Pinouts and motor hooks stay in that sketch's README.md next to the .ino.

3 — Match settings, then connect

Open **Control panel** → settings (gear). **User type** = **Default**. **Connection** = Bluetooth, **Classic + Simple (text lines)**. Leave the screen, then tap the Bluetooth button on the module toolbar.

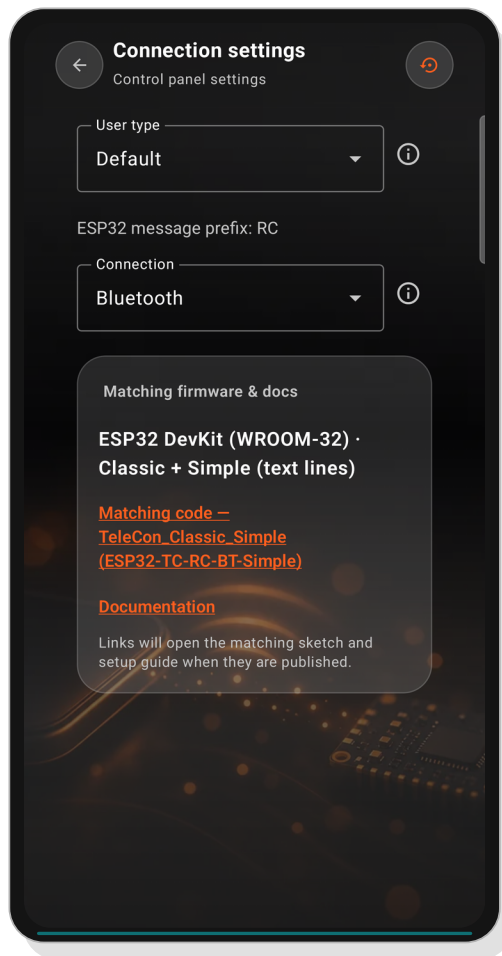


Figure 4: **S03.** Control panel settings. User type Default selected. Connection section showing Bluetooth + Classic + Simple (text lines). No Advanced options expanded.

Portrait, light theme, scrolled to User type ■ s03_settings_default.png

If Android asks to turn Bluetooth on, allow it.

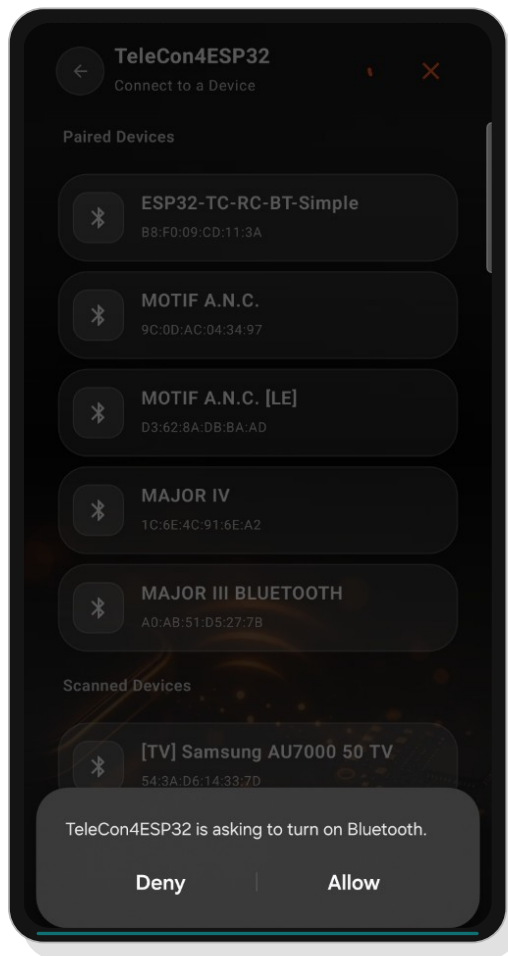


Figure 5: **S04.** Android Bluetooth enable sheet from the Connect screen (Deny / Allow). English UI on this capture.

System dialog, portrait ■ s04_permission.png

On **Connect to a Device**, tap refresh to scan. When ESP32-TC-RC-BT-Simple appears under **Scanned Devices**, stop the scan and tap that row. If it was paired before, use **Paired Devices**.

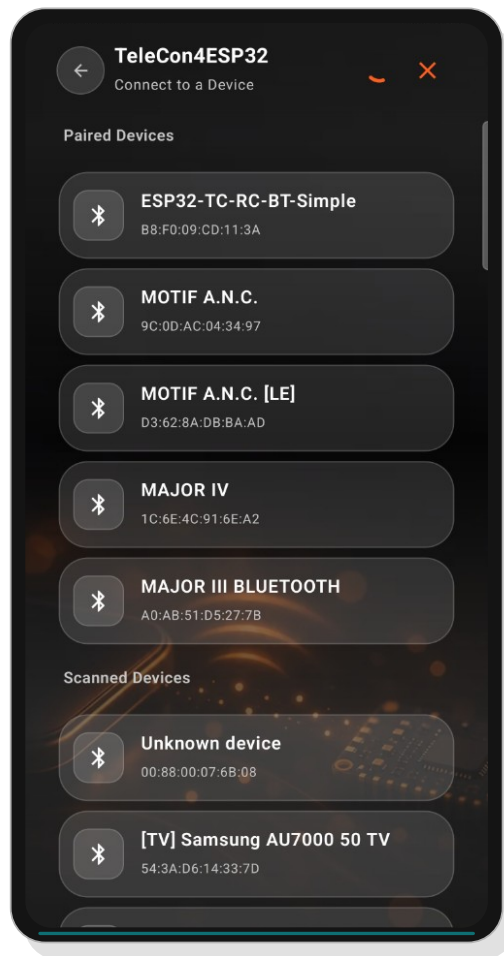


Figure 6: **S05**. Bluetooth screen with Paired Devices and Scanned Devices. ESP32-TC-RC-BT-Simple visible. English UI on this capture.

Portrait, Connect screen ■ s05_bluetooth_scan.png

Expected result. The app returns to the module. Status shows **Connected**. Home also shows Connected and highlights Control panel.

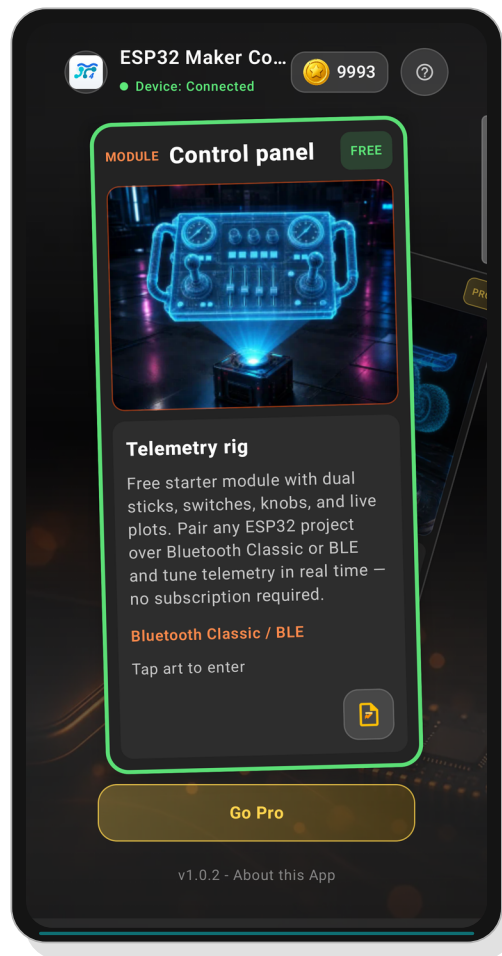


Figure 7: **S06.** Same Home layout as S01, but the top status reads Connected and the Control panel card is the highlighted recent module.

Portrait, light theme, connected ■ s06_home_connected.png

4 — Open Control panel and move a stick

Open **Control panel** (landscape). Drag the left stick. On the bench, official Simple firmware echoes motion into gauges and the plot (simulate mode, no extra wiring). In Serial Monitor you should see control traffic.



Figure 8: **S07**. Full Control panel HUD: both sticks, S1–S6, both knobs, side buttons, header gauges, seven-segment panels, LEDs, and the center plot. Connected (no “Not connected” banner). Optional: left stick off-center so the layout is obvious.

Landscape, light or the app’s plastic theme ■ s07_control_panel.png

If nothing moves, stay on this screen, confirm Connected, and check that settings still say Classic + Simple — see Section 12.

5 Home

Home is the launcher, not the live controller.

Control	What it does
Control panel / RC Vehicle	Opens that module. Expand a card for details, docs, or coin unlock.
Docs icon on a card	ZIP sketches and PDFs for that module.
Gold PRO	Subscription / upgrade.
Coin chip	Balance and pricing (when the coin economy is on).
Help (question mark)	Short FAQ for Home.
Title TeleCon4ESP32	About, privacy, and online documentation.

Browse Catalog appears only when more modules ship. Until then, tap a card on Home.

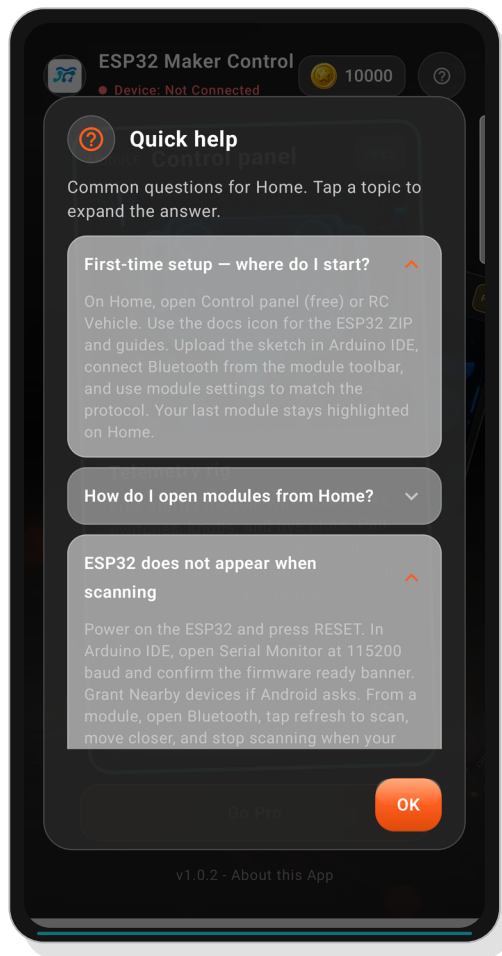


Figure 9: **S08.** Home with the Quick help dialog. Title Quick help, at least two topics expanded (first-time setup and ESP32 does not appear). English UI.

Portrait, Help dialog open ■ s08_home_help.png

6 Connect

Always connect *from the module* you will use (toolbar Bluetooth / Connect), not from a leftover system pairing alone. The module settings tell the app which link to open.

6.1 Bluetooth

Used by DevKit sketches: Classic Simple (Default), Classic Binary, BLE Binary (Advanced).

1. Allow Nearby devices (or Location on older Android).
2. Scan, then pick the board. The advertised name is only a label; the app connects by address.
3. Official Default name: **ESP32-TC-RC-BT-Simple**.

If a previous firmware is still bonded, Android may open the wrong profile. Forget the device in system Bluetooth settings and scan again.

6.2 Board Wi-Fi (SoftAP)

Used by CAM video and by some Advanced DevKit Wi-Fi builds. The phone leaves home Wi-Fi. No internet on that network is normal.

Default CAM network: SSID **TeleCon-RC-CAM-Starter**, password **telecon1234**. Then tap Connect in the module so control can use TCP 192.168.4.1:3333 (Role B) or stay on Bluetooth (Role A).



Figure 10: **S09.** System Wi-Fi picker with TeleCon-RC-CAM-Starter visible under Available networks (On). English system UI on this capture.

Portrait, Android Wi-Fi settings ▪ s09_wifi_join.png

Do not put SoftAP and Bluetooth on a single ESP32-CAM. Video CAM + control DevKit is the two-board (Role A) layout.

7 Control panel

Landscape remote for any ESP32 project. While this screen is open, sticks and knobs stream continuously.

7.1 Layout

Numbers match the callouts you will add on screenshot S07:

1. Header gauges — analog value and battery-style level from telemetry.
2. Seven-segment panels — large numbers (for example RPM). Titles and color come from Panel settings; values come from the board.
3. Status LEDs — eight lamps driven by a telemetry bitfield.
4. Joysticks — left and right. Spring vs hold is a setting.
5. Switches S1–S6 — three per side.
6. Rotary knobs — auxiliary 0–100% style values.
7. Action buttons — four momentary side buttons.
8. Center area — plot by Default; Camera, Radar, and Stick graph are extra unlocks.

7.2 Panel settings

From the module, open settings → initial switches, initial knobs, panel labels, plot channel names.

On the live Control panel HUD, **double-tap the stick** you want to configure (left or right). That opens the joystick sheet for that stick (Spring / Hold, axis orientation, stick shape).



Figure 11: **S10.** Control panel live HUD after double-tapping a stick: joystick settings sheet open (Spring / Hold, axis orientation, stick shape). English UI on this capture.

Landscape, Joystick Settings visible ■ s10_panel_settings.png

Center display (Plots / Stick / Camera / Radar) is a circular picker on the live HUD. Camera and Radar need coins or Pro.



Figure 12: **S11.** Control panel with the Center display sheet: Plots, Stick, Camera, Radar. Show which items are locked if running a free account.

Landscape, center-display picker open s11_center_picker.png

8 RC Vehicle

Pro driving cockpit. Unlock with subscription, coins, or the occasional explorer gift (short ad).

Two hardware layouts:

Role	Boards	What the phone does
Default B	One ESP32-CAM	Join CAM Wi-Fi. Video HTTP. Control on TCP :3333.
A overlay	CAM + DevKit	CAM Wi-Fi for video only. Sticks on DevKit Bluetooth.

One CAM — Default CAM video + DevKit Bluetooth Not one CAM for both SoftAP and Bluetooth

Steer center trim lives under Tune on the HUD (tickers → Check). The board stores the bias.

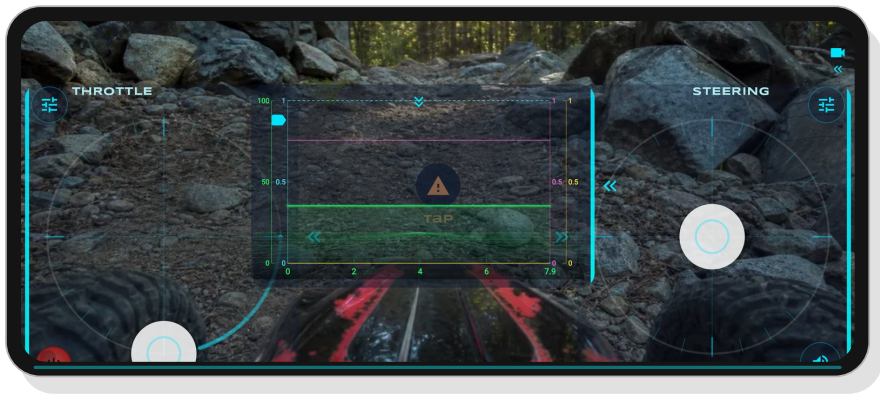


Figure 13: **S12.** RC Vehicle live screen: dual sticks, HUD telemetry, and the camera pane (live MJPEG or the in-app camera help card if the stream is idle). English UI. Prefer Role B Default with video online if you can.

Landscape HUD s12_rc_vehicle.jpg

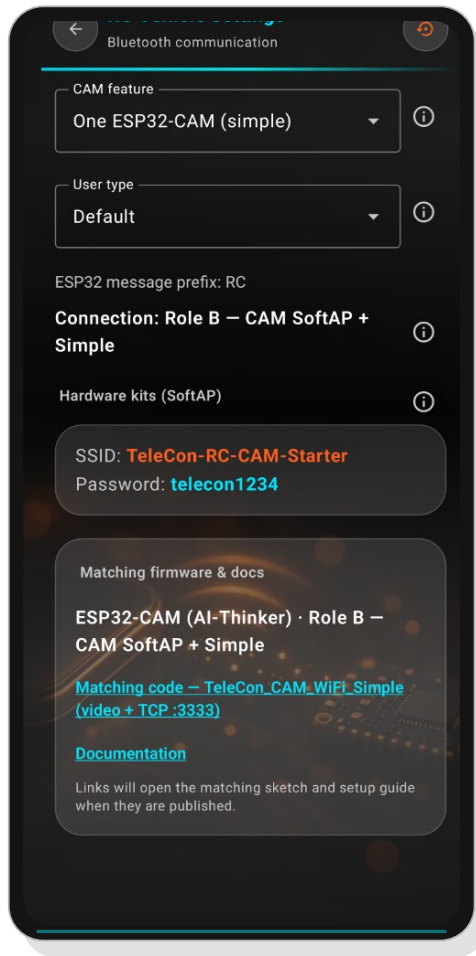


Figure 14: **S13**. RC Vehicle settings: CAM feature One ESP32-CAM (simple), User type Default, Role B SoftAP + Simple, SSID TeleCon-RC-CAM-Starter. English UI on this capture.

Portrait, Connection / CAM section ■ s13_vehicle_settings.png

9 Default vs Advanced

User type in module settings hides the scary options on purpose.

User type	Typical link
Default	DevKit Classic Simple, or one CAM SoftAP Simple, or Role A (video-only CAM + DevKit Classic Simple). Free.
Advanced	Classic Binary, BLE Binary, DevKit Wi-Fi Binary, CAM SoftAP Binary. Coins or Pro.

The firmware on the board must match the row you pick. After a change, disconnect and connect again. **Restore original defaults** puts that module back on the starter Default link.

10 Codes and About

- **Codes and Documents** — this PDF, a short fast guide, and the ZIP that matches the board + mode you selected in settings.
- **About** — version, privacy policy, and the hosted documentation site.

11 Coins and Pro

Control panel stays free on Default Simple. **RC Vehicle** entry, Advanced links, and extra center views (Camera, Radar, session CSV) use coins or a Pro subscription.

The gold sparkle on a locked card is an explorer gift: a short ad unlocks that Pro module for a few hours.

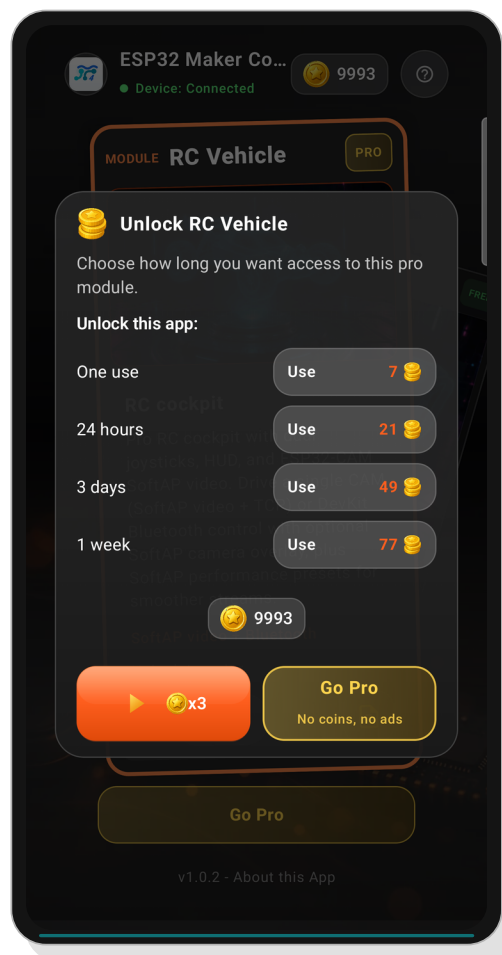


Figure 15: **S14**. Coin unlock dialog for RC Vehicle or Advanced settings. Show coin price options and the module name. English UI.

Portrait, unlock dialog ■ s14_coin_unlock.png

12 When it fails

Start from the symptom, not from the protocol.

You see	Try this
Board missing in scan	Power + RESET. Serial ready banner at 115200. Nearby devices allowed. Move closer. Stop scan when it appears.
Connect fails	Open Bluetooth from the module, not only from system settings. Forget a stale pair. Match Classic vs BLE to the sketch.
Home still Not connected	Connect inside the module. One phone only. Then Home updates.
Sticks / gauges dead	Stay on the live screen. Settings must match the sketch (Default Classic Simple for the first session). Reflash if Serial never shows ready.
“Couldn’t connect” / handshake timeout	The phone opened a pipe but the board did not answer OK in time. Same mode on both sides; update the sketch if it ignores the start request.
Protocol doesn’t match	Settings say Simple and the board speaks Binary (or the reverse). Change settings or flash the matching ZIP.
No camera	Join the CAM SSID first. Password telecon1234 . Role A: video on CAM Wi-Fi, sticks on DevKit Bluetooth — not both radios on one CAM.
Motors keep running after drop	Official sketches stop. If you wrote firmware, add fail-safe before any vehicle test.

Xiaomi / POCO on SoftAP: turn off mobile data if the TCP link dies, and keep TeleCon in front. VPN off.

13 Where to go next

- Sketch **README** next to the `.ino` — pins, SSID, BT name, **YOUR CODE HERE**.
- **Control Protocol PDF** — if you write your own firmware.
- In-app Help — short reminders on the phone.

A Screenshot checklist

Replace each yellow box with the file named in the header. Prefer English UI for the English PDF (Spanish UI for the Spanish PDF), light theme, cropped to the app (except S04 and S09, which are system sheets).

ID	File	Shot
S01	s01_home.png	Home, not connected, both module cards.
S02	s02_codes.png	Codes & Documents, Control panel.
S02b	s01_serial.png	Arduino Serial Monitor, Classic Simple banner.
S03	s03_settings_default.png	Settings: Default + Classic Simple.
S04	s04_permission.png	Android Bluetooth / Nearby dialog.
S05	s05_bluetooth_scan.png	Scan list with ESP32-TC-RC-BT-Simple.
S06	s06_home_connected.png	Home, Connected.
S07	s07_control_panel.png	Control panel landscape HUD.
S08	s08_home_help.png	Home Quick help dialog.
S09	s09_wifi_join.png	System Wi-Fi joining CAM SoftAP.
S10	s10_panel_settings.png	Joystick Spring / Hold.
S11	s11_center_picker.png	Center display picker.
S12	s12_rc_vehicle.jpg	RC Vehicle HUD + camera.
S13	s13_vehicle_settings.png	RC Vehicle Default CAM settings.
S14	s14_coin_unlock.png	Coin unlock dialog.